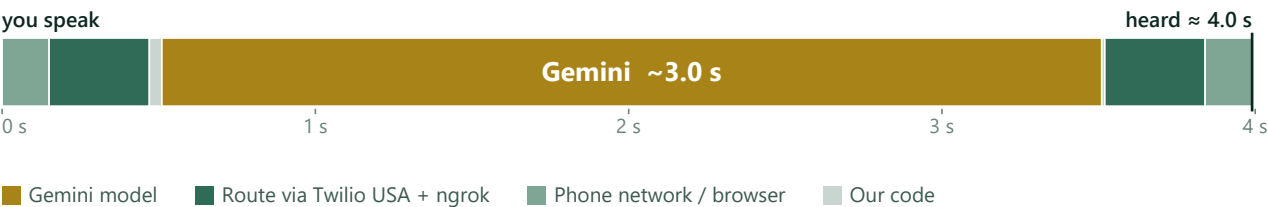


Where the translation delay comes from

From the moment someone speaks to the moment the other person hears the translation takes about **4 seconds**. This page shows which part of the journey takes that time. Measured on 6 October 2026 with Gemini 3.5 Live Translate.

75% of the delay is the **Gemini model itself**. It needs about 3 seconds to hear, translate and speak each phrase. Everything else together adds about 1 second.

One sentence, start to finish (to scale)



Each part of the journey

Part	What happens	Time	How we know
Gemini model	Listens, translates and speaks the phrase (Google's servers)	~3.0 s	Measured: 3.08 s and 3.19 s
Route via Twilio USA + ngrok	Calls are processed in Twilio's Virginia region, then tunnelled to our PC in India, and back again	~0.5–0.8 s	From call logs (US-East connections, 0.75 s to connect)
Phone network / browser	Mobile network and browser audio in and out	~0.2–0.4 s	Typical figure
Our code	Audio conversion, 100 ms batching, sending to Google	< 0.1 s	Measured: 0.5 ms conversion, 38 ms to Google
Total		~3.8–4.3 s	

Why long sentences don't change this: Gemini translates while the person is still talking, so the translation runs about 3 s behind the speaker for the whole sentence. In our test the lag was 3.08 s at the start of the speech and 2.99 s at the end.

What we can do about it

- **Gemini's ~3 s cannot be tuned.** Google offers no speed setting for this model. Going much faster means testing a different model.
- **The route can be shortened by ~0.5 s.** Running the backend on a cloud server next to Twilio (US-East) removes the trips through ngrok and India.
- **It can feel faster.** Starting translators while the phone rings, stopping echo from cutting translations off, and playing the speaker's voice quietly under the translation all make the wait less noticeable.

Method: real Hindi and Tamil speech streamed through our own translation code in real time as 8 kHz phone audio, timestamping each stage; route figures from backend logs of demo calls on 5–6 October 2026. Gemini figures are for gemini-3.5-live-translate-preview.